

# HOW TO NAME UR DOGS IN 100 YRS

## ATTEMPT

I want to explore what people in different parts of New York City might name their dogs 100 years from now. The project treats dog names as a cultural population: names are copied and move between areas, disappear, and occasionally mutate into new names.

The idea is to get plausible speculative names based on patterns already present in the city’s dog-licensing records. Its final interface could ask for a ZIP code and return a locally evolved name for a future dog. It might also go backward to see what was trendy historically.

## DATA

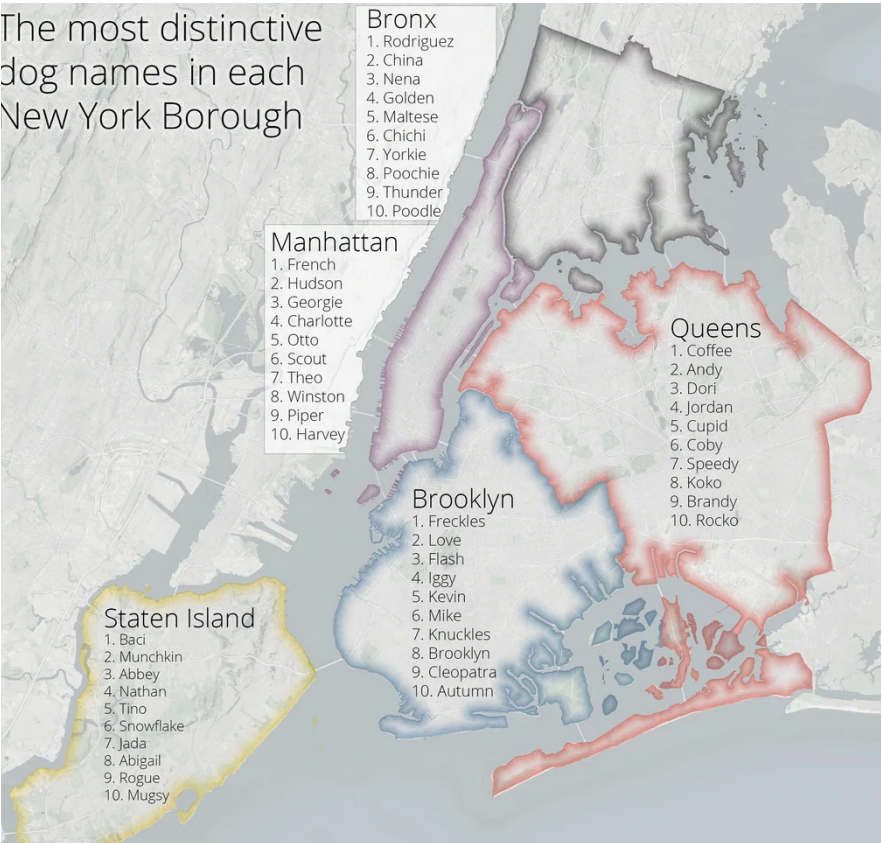
The NYC Dog Licensing Dataset is produced by the New York City Department of Health and Mental Hygiene from its Dog Licensing System. It was made public in 2017 and is updated periodically for administrative and licensing purposes. It contains 819,323 rows and eight columns, with license issue dates from September 2014 through December 2025.

Useful columns include: Animalnames, BreedName, ZipCode, AnimalBirthYear, LicenseIssuedDate.

A row represents a license record, not necessarily one unique dog. Renewals create repeated records. I would remove exact duplicates, remove placeholder names such as “UNKNOWN,” and approximately identify repeated dogs using name, gender, birth year, breed, and ZIP code.

ZIP codes provide a neighborhood-scale geography, although they are postal zones rather than official neighborhood boundaries.

The most distinctive dog names in each New York Borough



## Columns (8)

| Column Name        | Description                                         | API Field Name     |
|--------------------|-----------------------------------------------------|--------------------|
| AnimalName         | User-provided dog name (unless specified otherwise) | animalname         |
| AnimalGender       | M (Male) or F (Female) dog gender                   | animalgender       |
| AnimalBirthYear    | Year dog was born                                   | animalbirth        |
| BreedName          | Dog breed                                           | breedname          |
| ZipCode            | Owner zip code                                      | zipcode            |
| LicenseIssuedDate  | Date the dog license was issued                     | licenseissueddate  |
| LicenseExpiredDate | Date the dog license expires                        | licenseexpireddate |
| Extract Year       |                                                     |                    |

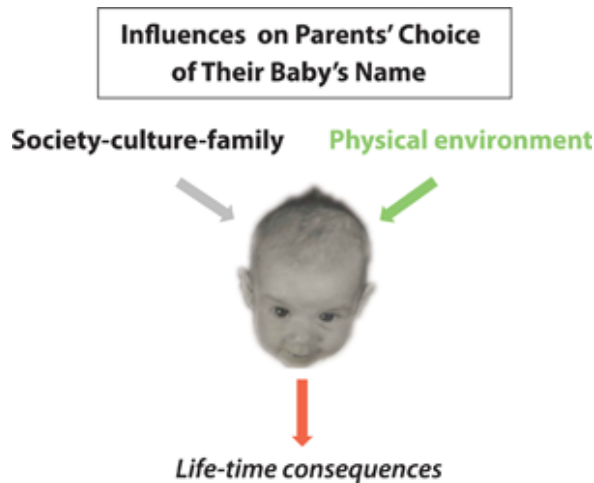
## RESEARCH

Baby-name researchers have modeled names as cultural traits that change through copying and innovating. Gureckis and Goldstone's "How You Named Your Child" uses recent popularity and momentum to model future naming choices.

Hahn and Bentley's random-drift model of baby names offers an especially useful precedent. In their model, each new name is either copied from the existing population or introduced through innovation. This produces the rise and disappearance of fashionable names without claiming to know why an individual selected a name.

The name generator is based on a character-level language model. This method originates in probabilistic models of letter sequences, such as Claude Shannon's A Mathematical Theory of Communication, in which the probability of a letter depends on preceding letters. A practical example is PyTorch's character-level name generator, which produces names one character at a time until it reaches an end marker.

```
python3 dog_name_generator.py \
  /Users/yinuoliu/Downloads/
  NYC_Dog_Licensing_Data-set_20260930.csv \
  --zip 11226 \
  --start-year 2018 \
  --end-year 2022 \
  --count 10 \
  --seed 2126
```



## STRATING RECORDS

Animal Name + ZIP code

**Name Generator**

Dafne Carney  
Micheline Manns  
Fioralba O'Brien  
Elnora Rolinson  
Norina McGarvey  
Gemma Hinder  
Simonetta Sinatra  
Vincenza Childs  
Leonora Doone  
Ernesta Fagon  
Letizia Crosse  
Vanna Moroney  
Mariella Keaveney  
Bettina Blaxley  
Santa Gilson  
Immacolata Tullett  
Orsola Kyd  
Rossella Werner  
Fiammetta Durrant  
Drina Mitchard  
Fulvia Tyrer  
Antonietta Finch  
Nerezza Painter  
Beatrice Johnston  
Giustina Ballard

1    Generate 44 names    500

Generate Names

Gender: ☐ Male ☒ Female ☐ Either

☐ Attempt all iteration  
☐ Double-barrelled surnames  
☐ Forenames use initials only

Number of forenames/initials: 1

Set forename:

Set surname:

Low    Obscurity Level    All

Name lists to include in name generation:

| List                            | Include                             |
|---------------------------------|-------------------------------------|
| ♀ Hindi and Indian Names (Fe... | <input type="checkbox"/>            |
| ♂ Hindi and Indian Names (Male) | <input type="checkbox"/>            |
| ♀ Italian Names (Female)        | <input checked="" type="checkbox"/> |
| ♀ Japanese Names (Female)       | <input type="checkbox"/>            |

+ -

## RULE

At each annual step, the simulator creates a new cohort of dogs in every ZIP code. The size of the cohort equals that ZIP code's average annual number of dogs born between 2018 and 2022.

For each new dog, the model generates a random number from 1 to 100:

1 to 70: Copy a name used in the same ZIP during the previous five years

71 to 95: Copy a name used elsewhere in NYC during the previous five years

96 to 100: Invent a new name

Existing names are selected in proportion to their popularity. A common local name is therefore more likely to be copied than a rare one.

When a new name is generated, the model examines names used in that ZIP during the preceding five years. It records:

- Common starting letters
- Which letters commonly follow one another
- Typical name lengths
- Common name endings

The generator selects a starting letter and adds one letter at a time according to these observed patterns. It stops when it reaches a length sampled from the local name-length distribution.

Training scope: ZIP 11226  
Usable approximate dog records: 1,233  
Distinct names: 897

Generated candidates:

Bigh  
Hed  
Samurisy  
Rhed  
Wingto

Letter-by-letter trace for Bigh:

|      |    |       |                   |
|------|----|-------|-------------------|
| '^^' | -> | 'B'   | probability=10.9% |
| '^B' | -> | 'I'   | probability=11.2% |
| 'BI' | -> | 'G'   | probability=8.3%  |
| 'IG' | -> | 'H'   | probability=17.6% |
| 'GH' | -> | 'END' | probability=20.0% |

## MY RUN

The run advances in one-year steps from 2023 to 2126, producing 104 future naming cohorts. Each dog's choice between local copying, citywide copying, and invention is left to chance. The selection of parent names for an invented name is also random.

I would repeat the full simulation 100 times. This number is my choice: it is large enough to show a range of possible futures without implying that one run is the correct future.

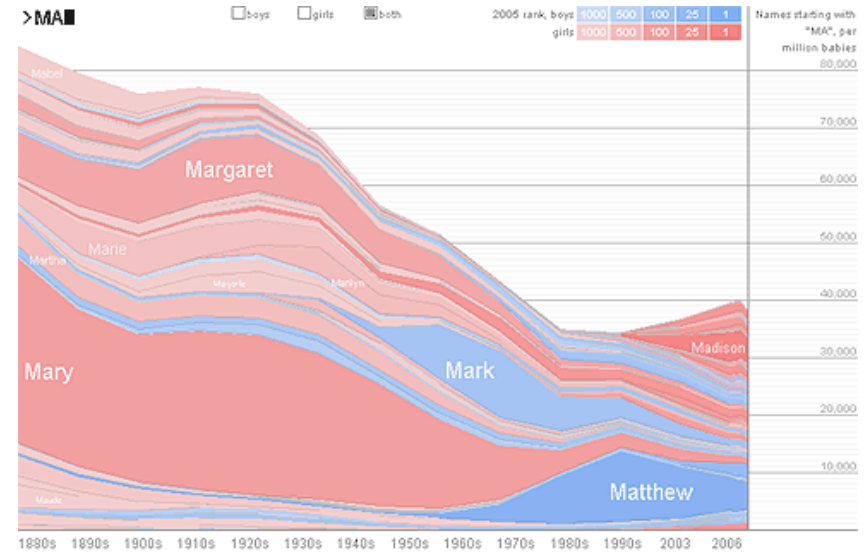
The final interface could ask:

What is your ZIP code? 11226 (if ZIP code still exists)

What year is your dog from? 2126

It would then sample a name from that ZIP code's simulated 2126 naming population.

|                      |                      |                       |                       |                         |                         |
|----------------------|----------------------|-----------------------|-----------------------|-------------------------|-------------------------|
| 1 Mariam Frazier     | 2 Callum Barker      | 3 Remington Santos    | 4 Walker Ayala        | 5 Blair Coleman         | 6 Micah Gutierrez       |
| 7 Savannah Andersen  | 8 Allstair Corona    | 9 Marianna Eaton      | 10 Leighton Andrade   | 11 Emmy Vega            | 12 Aidan Norman         |
| 13 Malani Espinoza   | 14 Dallas Chen       | 15 Valerin Benjamin   | 16 Kyro White         | 17 Layla Chan           | 18 Frank Norris         |
| 19 Arielle Meyer     | 20 Tristan Rodgers   | 21 Selah Dunlap       | 22 Aries McCarthy     | 23 Kira Blankenship     | 24 Ernesto Ramsey       |
| 25 Lyric Howard      | 26 Jeremiah Padilla  | 27 Maggie Whitaker    | 28 Keith Fitzpatrick  | 29 Annabella Pugh       | 30 Judson Rush          |
| 31 Maleah Ibarra     | 32 Asa Booker        | 33 Nataly Jaramillo   | 34 Riggs Cummings     | 35 Nylah Neal           | 36 Kane Leal            |
| 37 Murphy Cohen      | 38 Killian Gordon    | 39 Taylor Richmond    | 40 Mordechai Hester   | 41 Zendaya Barrett      | 42 Angelo Sutton        |
| 43 Izabella Wiggins  | 44 Azariah Hunter    | 45 Khloe Tang         | 46 Rogelio Pollard    | 47 Marisol Costa        | 48 Kenji Alfaro         |
| 49 Yasmin Mendez     | 50 Arthur Hill       | 51 Hannah McClure     | 52 Reese Clayton      | 53 Saige Scott          | 54 Leo Berry            |
| 55 Annabelle Perry   | 56 Waylon Levy       | 57 Flora Cooper       | 58 Jonathan Barry     | 59 Waverly Gilmore      | 60 Vihaan Serrano       |
| 61 Allie Dyer        | 62 Atreus Stephens   | 63 Millie Rice        | 64 Graham Mueller     | 65 Imani Magana         | 66 Rey Nunez            |
| 67 Mya Carey         | 68 Watson Garner     | 69 Jacqueline Haley   | 70 Lelf Gibson        | 71 Eden Gould           | 72 Blaine Christensen   |
| 73 Carmen Wyatt      | 74 Sam Chandler      | 75 Viviana Barr       | 76 Harley Larsen      | 77 Xiomara Escobar      | 78 Zachariah Fitzgerald |
| 79 Marlee Melendez   | 80 Nikolas Bean      | 81 Jenesis Combs      | 82 Ahmad Skinner      | 83 Mara Ryan            | 84 Timothy Patterson    |
| 85 Kaylee Glover     | 86 Mack Sanders      | 87 Everleigh Townsend | 88 Alexis Sexton      | 89 Ellen Henson         | 90 Bellamy Dudley       |
| 91 Hadleigh David    | 92 Alonso Horne      | 93 Marlowe Drake      | 94 Jalen Howard       | 95 Sophie Robertson     | 96 Emiliano McMahon     |
| 97 Belen Glenn       | 98 Zaid Church       | 99 Ayleen Scott       | 100 Leo Buckley       | 101 Theodora Santos     | 102 Walker Steele       |
| 103 Rylie Quintana   | 104 Kelvin Crosby    | 105 Kelly Underwood   | 106 Reece Christian   | 107 Anahi Petersen      | 108 Samson Hull         |
| 109 Andi Schmidt     | 110 Zayden English   | 111 Kelly Ross        | 112 Wesley Walls      | 113 Lilianna Blackwell  | 114 Marcellus Baxter    |
| 115 Lara Wiley       | 116 Mathew Lucas     | 117 Phoenix Villegas  | 118 Clyde Potts       | 119 Ellison Horton      | 120 Garrett Kaur        |
| 121 Holland Schmitt  | 122 Murphy Wilkins   | 123 Amalia Everett    | 124 Camilo Richard    | 125 Davina Weeks        | 126 Anders Blankenship  |
| 127 Rosalee Landry   | 128 Jaxx Hancock     | 129 Katelyn Woodard   | 130 Westley Armstrong | 131 Presley Flynn       | 132 Kannon Frazier      |
| 133 Octavia Frank    | 134 Braylen Ware     | 135 Eileen Daniels    | 136 Xander Lee        | 137 Scarlett Fitzgerald | 138 Peyton Murphy       |
| 139 Bella Kline      | 140 Ramon Bishop     | 141 Brooklynn Holland | 142 Brady Hendricks   | 143 Dani Bell           | 144 Emmett Erickson     |
| 145 Sabrina Swanson  | 146 Hugo Church      | 147 Ayleen Carr       | 148 Kash Drake        | 149 Jayleen Gray        | 150 Nicholas Estes      |
| 151 Brittany Hanson  | 152 Khalil Lowery    | 153 Estelle Perkins   | 154 Kyrie Lin         | 155 Makenna Kemp        | 156 Melvin Sampson      |
| 157 Melani Cobb      | 158 Raphael O'Neill  | 159 Kenna Sawyer      | 160 Jefferson Stevens | 161 Katherine Holloway  | 162 Sutton O'Donnell    |
| 163 Bellamy Richmond | 164 Mordechai Vargas | 165 Andrea Cunningham | 166 Alejandro Donovan | 167 Azariah Murray      | 168 Ashton Bonilla      |
| 169 Romina Travis    | 170 Willie Yang      | 171 Angelina Nixon    | 172 Cory Person       | 173 Dylan Orozco        | 174 Keanu Cano          |
| 175 Egypt House      | 176 Yehuda Parsons   | 177 Maia Higgins      | 178 Sterling Carter   | 179 Lucy Patrick        | 180 Derrick Sanders     |

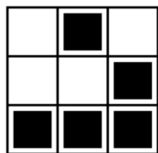


<https://1000randomnames.com/>

This page generates random names using the one thousand most common last names from the 2010 US census; the one thousand most common 2020 female given names from SSA; as well as the one thousand most common 2020 male given names from SSA. First and last names are combined randomly when the page loads. Odd-numbered first names are female and even-numbered first names are male. Duplicate first name-last name combinations are improbable since the page can produce 2M different names; however no duplicate check is performed, since it is possible for real names to be identical as well.

# Procedural Name Generator

Generate original names with Markov chains



MARKOV NAMEGEN

## WHAT IT CAN'T SEE

- intention behind the names
- Owner ethnicity, language, age, income, or social networks.

- Names borrowed from languages or cultures not represented in the current data.
- Dogs that were never licensed.
- Whether a dog was renamed, adopted, or moved after being named.
- Whether a generated name is pronounceable, meaningful, offensive, or culturally appropriate.